

ARYAN RAJENDRA SUTHAR

Software Engineer | Backend, Cloud & Distributed Systems

📍 Arizona, USA (Open to Relocation) 📞 813-279-8780 ✉️ aryansuthar6701@gmail.com [LinkedIn](#) [GitHub](#) [Portfolio](#)

PROFESSIONAL SUMMARY

Backend and distributed systems engineer with 4+ years delivering high-throughput microservices platforms — most recently architecting a 2M+ daily-transaction event pipeline at Avnet using Java, Spring Boot, Apache Kafka, and Kubernetes (AKS). Proven ability to cut infrastructure costs (27%), reduce API latency (38%), and boost data ingestion throughput (3.2×) across AWS and Azure environments. Specializes in event-driven architecture, cloud-native deployments, and AI-integrated search pipelines (LangChain, RAG, vector databases). Currently completing an MS in Software Engineering at Arizona State University (May 2026). Seeking senior backend or full-stack roles where deep systems expertise meets applied AI/ML.

TECHNICAL SKILLS

Languages: Java, Python, TypeScript, JavaScript, SQL, C#, Bash, Go (learning)

Backend & APIs: Spring Boot, FastAPI, Django, Node.js, Express.js, .NET Core, REST, gRPC, GraphQL, OAuth2, JWT

Frontend: React, Next.js, TypeScript, HTML5, CSS3, Component-Based Architecture

Cloud & Infrastructure: AWS (EC2, Lambda, S3, SNS, SES) · Azure (AKS, Functions, Service Bus, Blob Storage) · Docker · Kubernetes · Terraform (learning) · Serverless · Microservices

Event Streaming: Apache Kafka, Event-Driven Architecture, Message Queues, Stream Processing, Redis Pub/Sub

Databases & Caching: PostgreSQL, MySQL, MongoDB, DynamoDB, Redis, Elasticsearch (learning) · Query Optimization · Indexing

Observability & DevOps: GitHub Actions, CI/CD Pipelines, Prometheus (learning), Grafana (learning), Datadog, MLflow

AI / ML & GenAI: LangChain, RAG Pipelines, Vector Databases, Embeddings, LLMs, PyTorch, TensorFlow, NLP, Computer Vision

Testing & Quality: JUnit, Pytest, Jest, TDD, Unit Testing, Integration Testing, CI-based Test Automation

PROFESSIONAL EXPERIENCE

Software Engineer (Contract) | [Avnet](#) | Arizona, USA | Dec 2025 – Present

- Architected a high-throughput microservices platform using Java, Spring Boot, and Apache Kafka — processing 2M+ daily transactions across Kubernetes (AKS) clusters, improving system scalability by 45% under peak load
- Engineered low-latency RESTful and gRPC APIs with Node.js and .NET Core, reducing average response time by 38% and enabling seamless cross-service communication across 10+ distributed microservices
- Optimized cloud-native deployments on AWS (EC2, Lambda, S3) and Azure (AKS, Service Bus) using autoscaling policies and cost-allocation analysis — cutting monthly infrastructure spend by 27% while maintaining 99.98% uptime
- Implemented event-driven data pipelines with Kafka and asynchronous processing, increasing data ingestion throughput by 3.2× and eliminating bottlenecks in real-time analytics workflows
- Designed advanced PostgreSQL and MongoDB schemas with optimized indexing strategies, improving query performance by 41% and reducing data retrieval latency for large-scale datasets
- Shipped production RAG search pipeline using LangChain, vector embeddings, and Azure Cognitive Search — enhancing internal query relevance by 33% and reducing manual lookup time for analyst teams
- Collaborated with product and infrastructure teams to define API contracts and architecture standards across 3 cross-functional squads, accelerating feature delivery by ~20%

Software Engineer | [Orion Techno lab](#) | India | Jan 2021 – Jul 2024

- Developed scalable backend systems using Python, Django, and FastAPI — handling 500K+ daily API requests with 35% improvement in processing efficiency through caching and async I/O optimization
- Built responsive frontend applications with React and Next.js, boosting user engagement by 28% via optimized component rendering, lazy loading, and Core Web Vitals improvements
- Deployed containerized applications using Docker and Kubernetes, achieving zero-downtime blue/green releases and improving deployment frequency by 2.5× via GitHub Actions CI/CD pipelines
- Strengthened API security by implementing OAuth2 and JWT-based authentication with role-based access control, reducing unauthorized access incidents by 60% and achieving SOC 2 alignment
- Leveraged Redis distributed caching with TTL strategies, decreasing database load by 37% and improving P95 response time during high-concurrency traffic spikes
- Mentored 2 junior engineers through code reviews and pair-programming sessions, improving team defect detection rate by 25% and onboarding time by 30%
- Integrated TensorFlow and MLflow ML pipelines into product features, increasing predictive model accuracy by 22% and enabling data-driven personalization at scale

FEATURED PROJECTS

PhysioApp — AI-Powered Rehabilitation Platform — Next.js · FastAPI · PostgreSQL · Supabase · GitHub Actions | [GitHub ↗](#)

- Built full-stack physiotherapy platform with personalized exercise routines, multi-role auth (patient/therapist/admin), body metrics validation, and a 30-second rest timer system; deployed with GitHub Actions CI/CD on Vercel + Railway
- Designed normalized relational schema supporting 12+ entity relationships and analytics dashboard tracking weekly patient progress across multiple rehabilitation programs
- Integrated email support via SES and an FAQ/Help Center module; reduced onboarding friction by ~40% through guided multi-step wizard with real-time boundary validation

Distributed AI Simulation Platform — Python · LangChain · RAG · FastAPI · React · Vector DB | [GitHub ↗](#)

- AI-powered courtroom simulation platform using multi-phase prompt orchestration and RAG retrieval over legal document corpora for selfjustice.ca — enabling realistic legal practice with sub-2s response latency
- Architected event-driven backend with FastAPI and Redis pub/sub for real-time simulation state management across concurrent sessions

EmoFusion — Multimodal Emotion Detection System — Python · PyTorch · HuBERT · OpenCV · NLP · Late-Fusion Architecture | [GitHub ↗](#)

- Late-fusion deep learning system combining computer vision (facial landmark detection) and NLP (speech embeddings via HuBERT + 1D-CNN) for multimodal emotion classification across 7 categories
- Achieved 84% weighted F1-score on benchmark dataset; modular pipeline supports swapping vision/audio encoders independently for ablation studies

AI Knowledge Assistant — Python · LangChain · FastAPI · Redis · Vector Search · PostgreSQL | [GitHub ↗](#)

- Production-oriented RAG system with semantic vector search, Redis response caching (60% latency reduction), and document-grounded Q&A with source attribution — sub-300ms average query response
- Implemented hybrid retrieval (dense + sparse) with re-ranking for improved answer accuracy on domain-specific corpora

EDUCATION

Master of Science, Software Engineering | Aug 2024 – May 2026

Arizona State University, Tempe, AZ Capstone: PhysioApp — AI-Powered Physiotherapy Rehabilitation Platform

Relevant coursework: Distributed Systems, Software Architecture, ML Systems, Software Engineering Practicum

CERTIFICATIONS

- AWS Services Fundamentals for Beginners — Simplilearn
- Microsoft Azure AI Essentials Professional Certificate
- DevOps Professional Certificate — PagerDuty + LinkedIn Learning
- Atlassian Agile Project Management Professional Certificate
- AWS Certified Solutions Architect – Associate (In Progress)

ACHIEVEMENTS

- Solved 500+ algorithm problems on LeetCode — strong foundations in data structures, dynamic programming, and graph algorithms
- Contributed to PhysioApp capstone as lead developer for Industry Team 16 across 4 agile sprints (SER 517, ASU) — produced comprehensive technical documentation covering full system architecture and database schema
- Designed and maintained 10+ production-ready projects spanning AI/ML, distributed systems, and full-stack web development — available at [portfolio-pi-three-93.vercel.app](#)

References available upon request · [portfolio-pi-three-93.vercel.app](#)